

MONROE COUNTY AP, IN, USA

WMO: 724375

Lat: 39.146N

Lon: 86.616W

Elev: 257

StdP: 98.27

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 03893

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.1	-12.7	-20.7	0.6	-14.4	-17.9	0.8	-11.7	10.5	6.4	9.2	6.7	1.8	290	0.461

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	32.8	23.9	31.7	23.8	30.5	23.0	25.8	30.5	25.1	29.8	24.3	28.6	3.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.4	19.9	28.6	23.6	19.1	27.9	22.8	18.1	26.9	81.2	30.6	77.6	29.8	74.5	28.7	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.7	7.7	7.0	-20.6	34.5	3.9	2.3	-23.5	36.2	-25.7	37.6	-27.9	38.9	-30.8	40.6		
			-21.0	26.9	3.9	0.9	-23.8	27.6	-26.0	28.1	-28.2	28.7	-31.0	29.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.3	-1.0	0.8	6.5	12.8	17.9	22.4	23.7	23.1	19.5	13.2	6.8	1.4
	DBStd	10.14	6.91	6.53	6.12	5.14	4.45	3.24	2.96	2.78	3.97	4.96	5.42	6.23
	HDD10.0	1212	347	265	145	29	2	0	0	0	0	23	125	275
	HDD18.3	2791	601	492	371	177	64	7	1	2	33	173	346	526
	CDD10.0	2057	5	7	36	113	248	371	426	407	287	122	30	7
	CDD18.3	592	0	0	2	11	52	127	168	150	69	14	0	0
	CDH23.3	5396	0	0	15	101	442	1093	1575	1381	673	113	3	0
	CDH26.7	1810	0	0	1	12	108	366	581	490	231	23	0	0
Wind	WSAvg	2.9	3.6	3.5	3.6	3.6	2.9	2.5	2.1	1.9	2.0	2.6	3.2	3.3
Precipitation	PrecAvg	1121	71	65	92	105	127	104	117	95	89	77	93	83
	PrecMax	1668	229	154	214	259	271	265	300	219	259	203	224	189
	PrecMin	722	1	8	17	28	20	16	35	25	4	6	15	9
	PrecStd	206	49	34	51	51	67	54	60	52	65	47	46	38
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.7	20.2	25.3	28.0	31.2	32.9	35.9	34.6	33.3	29.2	23.0	18.9
		MCWB	15.1	15.2	16.8	18.0	22.3	23.7	24.5	23.8	21.9	21.5	16.5	15.6
	2%	DB	15.1	16.5	22.1	26.2	29.2	31.7	32.8	32.4	31.4	26.4	20.9	16.1
		MCWB	12.2	12.5	15.1	17.3	20.7	23.6	24.9	24.0	22.2	18.8	14.9	13.7
	5%	DB	12.4	13.7	19.1	24.1	27.8	30.5	31.4	31.1	29.4	23.9	17.9	13.2
		MCWB	10.4	10.4	13.3	16.1	20.0	22.8	24.6	23.4	21.7	17.6	13.6	10.9
	10%	DB	9.3	11.1	16.7	22.0	26.1	28.9	29.9	29.4	27.4	22.0	16.0	10.6
		MCWB	6.7	7.6	11.9	15.3	19.4	22.1	23.6	22.5	20.2	16.8	12.2	8.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.8	16.0	17.6	20.5	23.5	25.8	27.4	26.5	24.9	22.7	18.1	16.2
		MCDB	17.2	19.3	23.7	26.4	28.5	30.9	31.7	31.5	30.3	27.3	21.1	18.4
	2%	WB	13.1	13.5	16.1	18.8	22.4	24.9	26.3	25.5	23.6	21.0	16.2	13.7
		MCDB	14.7	15.6	20.8	23.6	27.5	29.8	30.7	30.5	28.6	24.6	19.1	15.5
	5%	WB	10.6	11.1	14.6	17.6	21.4	24.0	25.5	24.7	22.7	19.2	14.5	11.1
		MCDB	12.2	13.3	18.3	21.8	25.8	28.7	29.9	29.4	27.3	22.5	17.5	12.8
	10%	WB	6.9	7.5	12.6	16.3	20.4	23.2	24.6	23.8	21.6	17.4	12.4	8.3
		MCDB	8.7	10.3	16.0	20.4	24.5	27.4	28.7	27.9	25.4	20.6	15.3	10.1
Mean Daily Temperature Range	5% DB	MDBR	9.0	9.7	10.9	11.9	11.2	10.8	10.8	11.7	12.7	11.8	10.5	8.3
		MCDBR	11.2	12.7	13.9	14.2	12.9	12.0	12.0	13.2	14.8	14.0	13.4	10.6
		MCWBR	9.8	9.8	8.6	7.4	6.1	5.5	5.2	6.0	6.6	7.5	9.2	9.0
	5% WB	MCDBR	10.5	11.3	12.2	10.9	10.7	10.2	10.2	10.9	11.6	11.2	11.5	9.6
		MCWBR	9.8	10.0	8.3	7.0	5.7	5.2	5.2	5.8	5.9	6.6	9.1	8.9
Clear-Sky Solar Irradiance	taub	0.304	0.318	0.349	0.400	0.431	0.442	0.459	0.456	0.421	0.354	0.321	0.297	
	taud	2.464	2.399	2.386	2.248	2.207	2.212	2.163	2.160	2.242	2.440	2.478	2.491	
	Ebn at Noon	872	905	908	876	853	841	822	814	820	852	842	853	
	Edh at Noon	80	97	110	134	143	142	148	145	126	93	78	72	
All-Sky Solar Radiation	RadAvg	1.82	2.60	3.67	4.79	5.41	6.21	5.98	5.52	4.56	3.26	2.19	1.54	
	RadStd	0.20	0.31	0.33	0.34	0.50	0.38	0.36	0.33	0.37	0.33	0.24	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page